Poisson Distribution

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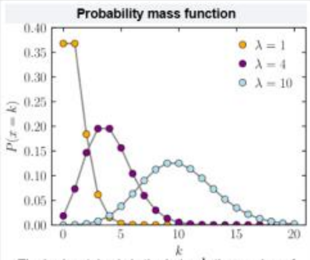
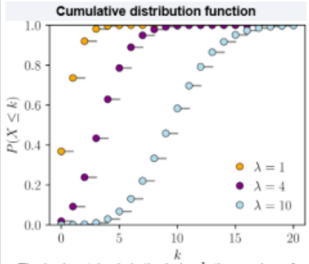
Poisson distribution



The horizontal axis is the index k, the number of occurrences. λ is the expected rate of occurrences. The vertical axis is the probability of k occurrences given λ. The function is defined only at integer values of k; the connecting lines are only guides for the eye.

Poisson distribution is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time if these events occur with a known constant mean rate and independently of the time since the last event.



The horizontal axis is the index k, the number of occurrences. The CDF is discontinuous at the integers of k and flat everywhere else because a variable that is Poisson distributed takes on only integer values.

① Discrete random variable (pmf)

Ex: Tossing a coin

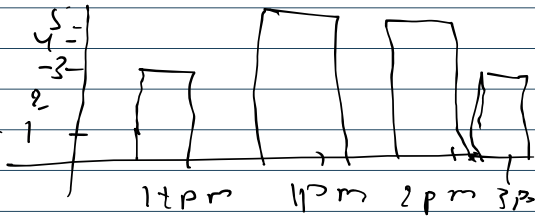
{H, T} => {0, 1}

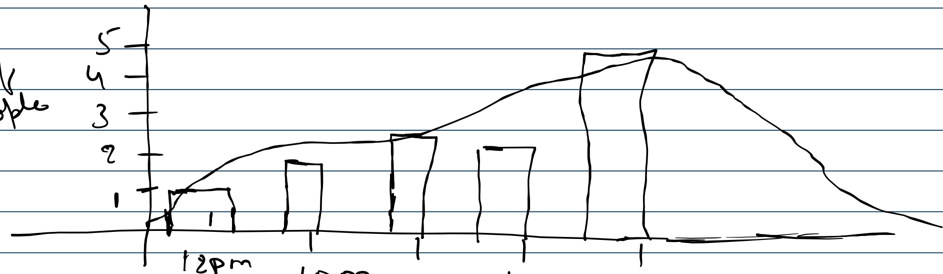
0, 1, 2, 3, 5, 6, ~~7~~

② Time Interval

Ex:- ① No. of visiting hospital

② No of Customers Visiting Bank





Area under Curve

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Probability

12pm - 5pm

$\lambda = 3 \Rightarrow$ Expected no. of Events occurs at Every time Interval

PmF

$\lambda = 3$ let us

$$p(x=5) \Rightarrow \frac{e^{-\lambda}\lambda^{x}}{x!} \Rightarrow \frac{e^{-3}\,3^{5}}{5!} \Rightarrow 0.101 \Rightarrow 10.1\%$$

Mean $= E(x) = \mu = \lambda \times t$

↓

Variance